

WikiContrib Final Report

Version 1.0 - Explainable Wikipedia Contributor-Impact Analysis

Field	Value
Project	WikiContrib - Wikipedia Contributor Impact
Author	Han Zhu (Han-ZhuAI)
Release	v1.0
Repository	github.com/Han-ZhuAI/wiki-contributor-impact
Evidence baseline	245 tests; 95.45% total coverage; Python 3.10 and 3.12 CI
Evaluation scope	Earliest 50 article and Talk revisions for Alan Turing, Kimchi and Python

Executive summary

Raw edit counts reward activity, not durable or collaborative impact. WikiContrib instead creates an auditable evidence chain from public MediaWiki revisions to four independent contributor axes: volume, additive work, persistence and discussion influence. The system preserves raw metrics, normalised features, policy weights and weighted terms so every rank can be reproduced and explained.

DELIVERY STATUS Core analytics, evaluation, tests, CI, JSON export, visual summaries and written documentation are complete. This written report is the selected submission explanation; a demo video is intentionally outside this delivery.

Key findings

- Balanced and one-axis-emphasis policies were compared across three real articles.
- Worst top-10 overlap remained between 0.80 and 0.90; minimum Spearman rho remained between 0.941 and 0.951.
- Automation-aware review changed the Alan Turing balanced winner from Conversion script to Raul654, while preserving all revisions in diff and provenance evidence.
- The results describe historical slices and ranking sensitivity; they do not establish causal influence or objective contributor quality.

1. Problem and design goals

Wikipedia articles are produced by contributors with different roles. A high edit count may reflect repeated small corrections, automated transformations or reversions, while a low-count contributor may supply durable prose or resolve a disputed design on the Talk page. A useful impact model must therefore separate evidence types and avoid treating one proxy as ground truth.

Naive signal	Failure mode	WikiContrib response
Edit count	Rewards save frequency and automation	Use volume, role, persistence and discussion separately
Bytes added	Cannot show whether text survives	Chain word-token provenance to the final analysed revision
Current text	Hides maintenance and reverted work	Retain revision-level diff and edit classification
Talk post count	Treats every post as equal	Build a directed reply graph and compute PageRank
Single score	Conceals value judgements	Expose axes, weights, weighted terms and sensitivity

System pipeline

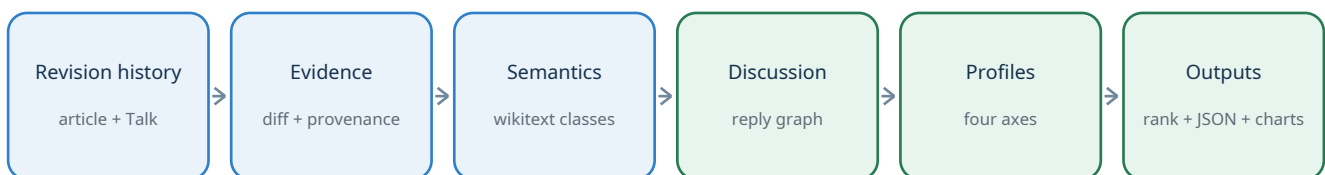


Figure 1. Deterministic path from public revision evidence to explainable outputs.

Contributor role taxonomy

Role	Observable evidence	Interpretation
Content creator	Net new prose and high surviving-word count	Adds durable article content
Maintainer	Reverts, trims, copy-edits and formatting work	Protects or improves existing content
Structural editor	Headings, references, templates, tables, categories and links	Changes article organisation or metadata
Coordinator	Signed Talk replies and reply-graph centrality	Participates in collaborative decisions
Automated candidate	Auditable username and comment indicators	Requires review; not an identity conclusion

2. Method

Token-level diff and provenance

Consecutive revisions are tokenised into Unicode word tokens and symbols. The diff records additions and removals, gross words, net words and churn. Provenance chaining assigns each surviving word token to its original contributor. Complete reverts and partial restorations reuse earlier ownership rather than assigning restored text to the restoring editor.

Four-axis contributor profile

Axis	Normalisation	Purpose
Volume	$\log(1 + \text{gross_words_u}) / \log(1 + \text{max_gross_words})$	Comparable scale of text touched
Additive	$\text{additive_edits_u} / \text{article_edits_u}$	Share of edits classified as additive
Persistence	$\log(1 + \text{surviving_words_u}) / \log(1 + \text{max_surviving_words})$	Durable original contribution
Discussion	$\text{pagerank_u} / \text{max_pagerank}$	Relative Talk reply-graph centrality

The composite score is the sum of each axis multiplied by its normalised policy weight. Default weights are 0.25 per axis. Any finite non-negative input scale is accepted and renormalised to one. Equal totals use username order as a deterministic tie-break.

Wikitext element classification

Every changed word is assigned to exactly one of seven locations: prose, heading, reference, template, table, category or link. Fixed precedence prevents nested markup from double-counting a word; comments and markup punctuation are excluded from word totals. These semantic deltas are reporting evidence rather than hidden score multipliers.

Talk reply graph

Signed and indented Talk posts are grouped by thread. A directed edge runs from the replier to the contributor being answered. Repeated replies increase edge weight; PageRank uses damping 0.85. Same-user article edits within 14 days after a Talk post are reported as temporal links, not as proof that the discussion caused the edit.

SCOPE RULE The safe CLI default analyses the earliest 500 revisions and labels the result `historical_slice=true`. Full-history persistence requires the explicit `--all-revisions` option.

3. Cross-article evaluation

EVALUATION SCOPE All reported cases use the earliest 50 article revisions and earliest 50 Talk revisions. They are historical slices, not current-article rankings.

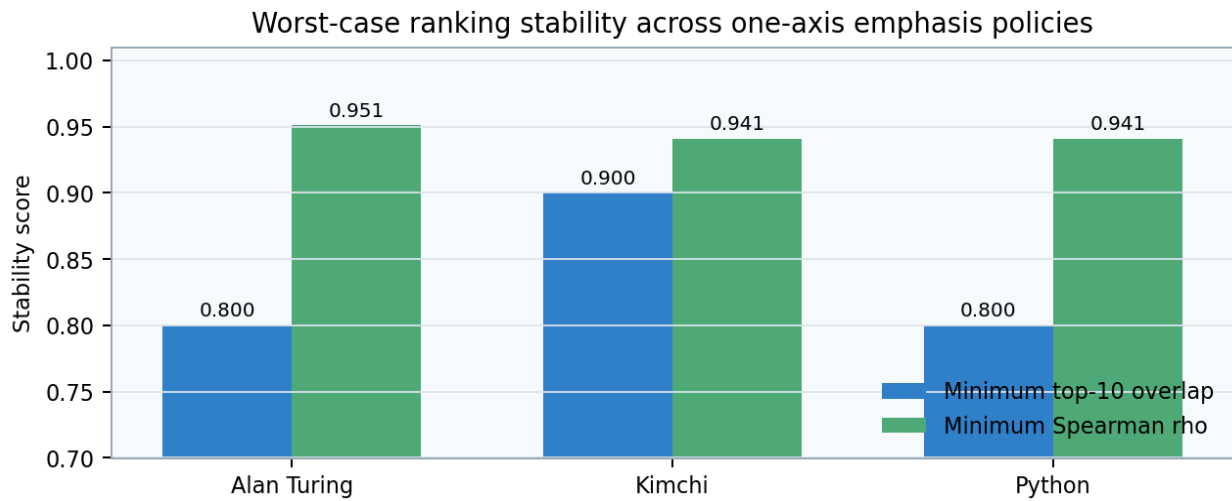


Figure 2. Minimum stability reached by any one-axis-emphasis policy.

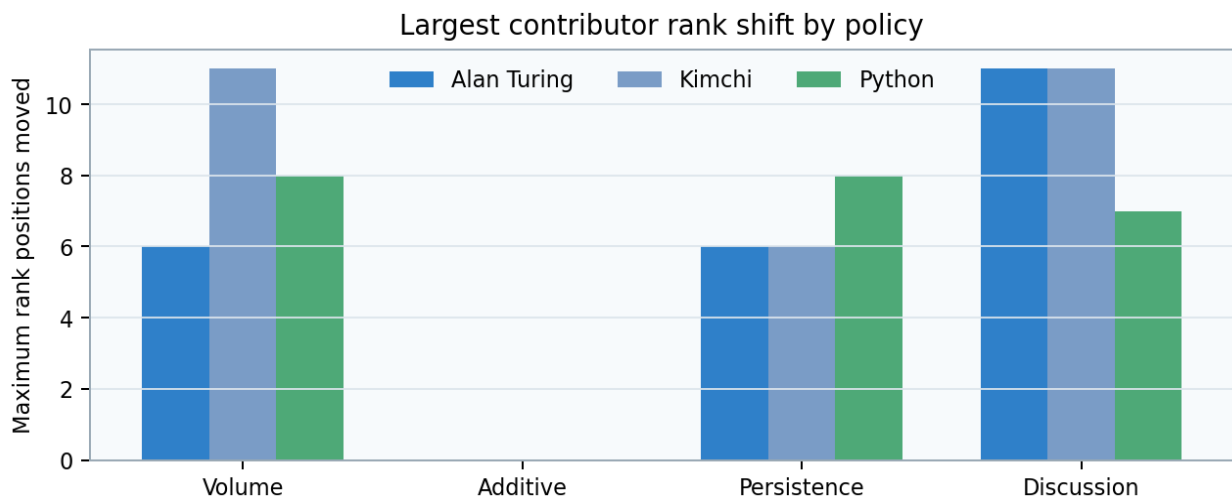


Figure 3. Maximum observed rank movement relative to the balanced policy.

Article	Contributors	Balanced winner	Filtered winner	Worst overlap	Min rho
Alan Turing	47	Conversion script	Raul654	0.800	0.951
Kimchi	41	Kokiri	Kokiri	0.900	0.941
Python	29	Fubar Obfusco	Fubar Obfusco	0.800	0.941

4. Case study and validation

Alan Turing historical slice

The balanced all-account policy ranks Conversion script first with a score of 0.750. The detector records high-confidence automation indicators for Conversion script and The Anomebot. Under the opt-in human-contributor comparison, Raul654 becomes the balanced winner with 0.728. Volume and persistence emphasis instead select Manning Bartlett after filtering, while discussion emphasis selects Raul654 before and after filtering.

Policy	Weights V/A/P/D	All-account winner	Filtered winner	Top-10 overlap	Max shift
Balanced	0.25/0.25/0.25/0.25	Conversion script	Raul654	1.000	0
Volume	0.40/0.20/0.20/0.20	Conversion script	Manning Bartlett	0.800	6
Additive	0.20/0.40/0.20/0.20	Conversion script	Raul654	1.000	0
Persistence	0.20/0.20/0.40/0.20	Conversion script	Manning Bartlett	0.900	6
Discussion	0.20/0.20/0.20/0.40	Raul654	Raul654	0.800	11

Why automation handling remains auditable

- Username bot/script markers form the high-confidence tier; comment-only evidence remains review-only.
- Candidate records retain confidence, reason codes and evidence revision IDs.
- Filtering changes only final ranking eligibility. It never deletes revisions or changes token provenance.
- The heuristic is not proof of bot status and should not be used to make identity claims without manual verification.

Verification baseline

Check	Verified result
Automated tests	245 passed
Coverage	95.45% total; CI threshold 95%
Python versions	3.10 and 3.12 GitHub Actions jobs pass
Regression focus	Exact reverts, partial restoration ownership, scoring and graph behaviour
Real-article benchmark	Alan Turing, Kimchi and Python

INTERPRETATION Sensitivity measures ranking stability, not objective accuracy. The evaluation supports reproducibility and policy transparency, not a ground-truth claim about who matters most.

5. Limitations, ethics and reproducibility

Limitations

- Revision caps can omit later article states and current contributors; every capped result must retain its scope warning.
- Token survival measures textual retention, not factual correctness, originality or reader value.
- Wikitext heuristics cannot resolve every malformed or deeply nested construct.
- Talk signatures, indentation and 14-day temporal links are incomplete collaboration proxies.
- PageRank measures graph centrality under a chosen edge direction and damping value; it is not authority or consensus quality.

Ethical use

WikiContrib analyses public revision histories, but public availability does not remove the need for careful interpretation. Rankings should be presented as model outputs under an explicit policy, never as a definitive judgement of a person. Anonymous and hidden contributors remain represented by source identifiers; the system does not attempt deanonymisation. Automation indicators remain review candidates rather than labels of intent or identity.

Reproduction

Install the package and run the same checks used by CI:

```
COMMANDS pip install -e ".[dev]"
ruff check src tests
python -m compileall -q src tests
pytest -q --cov=wikicontrib --cov-report=term-missing --cov-fail-under=95
```

Recreate the committed evaluation:

```
EVALUATE python -m wikicontrib evaluate "Alan Turing" "Kimchi" "Python (programming language)"
--max-revisions 50 --output-json report/evaluation.json --output-markdown report/evaluation.md
```

Conclusion

WikiContrib meets the intended v1.0 goal: a reproducible, explainable and auditable model that separates content creation, maintenance, survival and discussion influence. Its main contribution is not a universal ranking, but a transparent evidence pipeline in which scope, assumptions and value choices remain inspectable.

v1.0 delivery contents

- Python package and CLI with JSON and PNG outputs
- 245-test validation baseline and CI on Python 3.10 and 3.12
- Committed three-article sensitivity evaluation
- Final report, PRD, ER design, README and release notes